

ports 5GHz range and cannot be made to work at other lower ranges.

Unlike other Wi-Fi technologies, 802.11ac is not omni-directional and throws a single beam of signal from the access point (AP) to the device, and vice-versa. Even though this seems to be a positive feature that would allow fast streaming, signal tracking might be a challenge for devices in a large area or devices that have a weak signal detection strength.

Low-energy wireless communication

Last few years gave us the Bluetooth Low Energy (BLE) called Bluetooth Smart. Bluetooth Smart is not backward compatible with previous versions of Bluetooth protocol (classic), but Bluetooth 4.0 specification permits devices to either implement the BLE or the classic or both. Bluetooth 4.0 has supported Bluetooth Smart since over two years now, facilitating us with Bluetooth Smart Ready (compatible with both, classic and BLE) and Bluetooth Smart (BLE only) devices.

A new addition to this is the Bluetooth 4.1 specification that adds to the features of Bluetooth 4.0 by allowing the devices to support multiple roles at the same time. It allows Bluetooth Smart Ready products to act both as a peripheral and a hub at the same time. You would also be amazed to know that Bluetooth 4.1 can coexist with other wireless technologies at the same band (for example, the Wi-Fi 2.4GHz band) without much interference.

Nitin Gupta, lead engineer- μ Energy Applications, CSR, says that Bluetooth has been in use by mobile users since a long time now, so switching over to Bluetooth Smart is not a concern. Pointing out its essential features, Gupta says, "BLE has both a defined profile and an ability to define its profile, which is not ideally the case with other wireless technologies." Just to add to this, BLE has the ability to co-exist with other mobile wireless technologies on the same band with least interference, so there will be least chance of data loss. Further, it has limited discovery time due to which the

Other features of 802.11ac

- Runs on gigabytes speed which is three times greater than 802.11n.
- Few dead spots due to better range.
- Wide channels with high data rates of up to 1.3Gbps per radio.
- More bandwidth on mobile devices.
- Backward compatible with 802.11a and n at 5GHz band.

Where do you need 802.11ac?

- 802.11n already meets about 80 to 90 per cent of the bandwidth demands but 802.11ac still finds its presence important.
- In selective applications like telepresence and voice.
- In areas with continuous support with less expenditure.

How to deploy it?

- Replace 802.11n with dual-radio/dual-mode 802.11ac.
- Overlay 802.11ac only solution on an existing 802.11n network.
- Deploy hotspots with very high throughput (VHT) at places where client density is required or at places where localised higher data rates are required.

Other features of BLE based on Bluetooth 4.1

- Has both Dual Mode and Link Layer topology to support multiple roles and to detect and diagnose the signal efficiently.
- Allows updates of audio architecture for the wide-band speech.
- Generalised interlaced scanning and train nudging to ensure the operation in specific bands for successful device discovery and connectivity.
- Future communication through (logical link control and adaptation protocol) L2CAP dedicated channels is possible.
- Supports 802.11n media access control/physical layer (MAC/PHY) through the protocol adaptation layer update and enhances Bluetooth high-speed technology.

Other features of ZigBee Remote Control 2.0

- Replaces the IR technology with RF, providing new features like enhanced user interface (UI).
- Reduces problems related to technology, further reducing the service calls.
- Enables faster implementation, reducing the overall product development time.
- No range limit for use. Range extends in different rooms.

networks can be detected in less time.

With all the associated key features, Bluetooth 4.1 also has certain challenges when it comes to its application. The pairing of Bluetooth Smart devices (enabled with 4.1 specifications) with the long-distance feature allows eavesdropping on transactions, gives opportunities to track individuals, allows behaviour monitoring and enables spoofing of beacons. The other challenge is the cost and time invested to upgrade the infrastructure during the integration of the new Bluetooth technology. Gupta says, "Bluetooth Smart too has some design challenges when compared to other technologies. It has a lower footprint when

it comes to hardware implementation, but has a widely-adopted operating system (OS)."

The implementation of BLE requires efficient radio with the right antenna, as they all work on the same frequency domain, that is, 2.4GHz. If the right antenna is not used, there will be a lot of interference between the networks.

Bi-directional communication

The ZigBee standard operates on IEEE 802.15.4 (physical radio) specification, in the unlicensed bands (868MHz, 900MHz and 2.4GHz). However, there is a recent enhancement to this standard with the ZigBee Remote Control 2.0 that works on Radio Frequency for Consumer Electronics (RF4CE) specification. The RF4CE provides a simple networking layer and standard application layer to create a multiple-vendor operable solution to be used within a home.

The ZigBee Remote Control 2.0 uses faster and more reliable radio frequency (RF) technology as compared to infra-red (IR) for operation from greater distances. It consumes less power than IR and provides a fast and reliable bi-directional communication network, with an over-the-air update. Hence, it does not require to be connected with any external hardware for updates.

The ZigBee Remote Control 2.0 standard provides a new feature called 'Find My Remote' that allows other devices to detect and connect to the ZigBee module in smart homes. Since the new specification allows ZigBee to operate at